MIL-A-62219(AT)
7 June 1976

MILITARY SPECIFICATION

ALTERNATOR, 100 AMPERE RECTIFIED,
28 VOLTS D.C.

This specification is approved for use by U. S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the 28 Volt D.C. 100 ampere alternator with integral rectifier, used in one type of electrical generating system consisting of alternator-rectifier and solid state voltage regulator.

1.2 Classification. The alternator shall be of the following classes as specified (see 6.2).

Class A - Direct Driven.
Class B - Belt Driven.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION
Federal
NN-P-530 - Plywood; Flat Panel.
TT-P-104 - Paint (White Lead and Oil, Exterior, Ready-Mixed, White and Light Tints).
LLL-I-535 - Insulation Board, Thermal and Insulation Block, Thermal.

FSC-2920

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Military

MIL-P-116	-	Preservation, Methods of.
MIL-P-514	-	Plates; Identification, Transportation-Date and Blank.
MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification For.
MIL-B-11188	-	Batteries, Storage: Lead-Acid.
MIL-C-16173	-	Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-1275	-	Electrical Circuit, 28 Volts DC Voltage Characteristics in Military Vehicles.

Drawings

10947517	-	100 Ampere Alternator.
10947439	-	100 Ampere Regulator.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H 28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 First article. A first article of the alternators to be furnished under this specification shall be produced prior to the manufacture of the items in production quantities. Alternators furnished under this specification shall

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be products which are represented by a first article sample that has been tested and has passed the first article inspection specified herein (see 4.2 & 6.3). Approval of the first article sample by the activities concerned shall not relieve the contractor of his obligation to supply alternators in production that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

3.2 Materials. Materials shall be as specified herein, in applicable specifications and on applicable drawings. Materials not specifically designated shall be of good commercial quality. Materials used shall be suitable for use in the generator system operating over a temperature range from minus 65° to plus 250°F without any change in physical or dimensional properties that would result in calibration or operation of the generator system to fall outside of the limits specified herein.

3.3 Polarity. The alternator shall be designed to operate with either a positive or negative ground.

3.4 Design, construction and interchangeability. Unless otherwise specified, mounting and installation dimensions of the alternator fabricated in accordance with this specification shall have the mounting, shaft, and external dimensions in conformance with Drawing 10947517.

3.5 Standard parts. Military standard parts shall be used whenever possible.

3.6 Threaded surfaces. All threaded surfaces shall be in the inch system and shall conform to National Bureau of Standards Handbook H28.

3.7 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Separation of dissimilar metals, shall be accomplished by providing insulation between mating surfaces.

3.8 Performance.

3.8.1 Rated output. When tested as specified in 4.5.2, the alternator shall meet the minimum performance requirements specified for rated output on Drawing 10947517.

3.8.2 Voltage regulation. When tested in a system using Drawing 10947439 regulator, the output voltage of the generator system shall be established and maintained at 28 $\pm$ 0.5 volts under the load, speed and temperature conditions indicated.

3.8.3 Voltage characteristics.

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3.8.3.1 Load switching. When tested as specified in 4.5.3 the generator system shall not exceed the voltage surge requirements of MIL-STD-1275 and shall withstand the transient energy pulse generated without physical damage or impairment of performance. After the test, the generator system shall duplicate the rated output performance curve of Drawing 10947517.

3.8.3.2 High voltage transients. When tested as specified in 4.5.16, the alternator transient requirements shall fall within the transient characteristics of MIL-STD-1275.

3.8.4 Endurance. After having been subjected to the endurance test specified in 4.5.4, the generating system shall meet the requirements of 3.8.1 and 3.8.2. There shall be no servicing, adjustment or replacement of parts except that if brushes are used, one brush change will be permitted during the test.

3.8.5 Overspeed. After having been subjected to the test specified in 4.5.9, the generating system shall duplicate the rated output performance curve of Drawing 10947517. At the conclusion of the test, the alternator shall be visually examined. The alternator is required to withstand this test without any evidence of electrical or mechanical damage.

3.8.6 High temperature. After having been subjected to the test specified in 4.5.5 for 200 hours, the generating system shall duplicate the rated output performance curve of Drawing 10947517.

3.8.7 Low temperature. After having been subjected to the test specified in 4.5.6, the generating system shall duplicate the rated output performance curve of Drawing 10947517.

3.8.8 Shock. After having been subjected to the test specified in 4.5.7, the alternator shall show no physical damage and the generating system shall duplicate the rated output performance curve of Drawing 10947517.

3.8.9 Vibration. After having been subjected to the test specified in 4.5.8, the alternator shall show no physical damage and the generating system shall duplicate the rated output performance curve of Drawing 10947517.

3.8.10 Submersion. After the alternator has been subjected to the test specified in 4.5.13, the generating system shall duplicate the rated output performance curve of Drawing 10947517. Any sealed compartment shall contain not more than 10 cubic centimeters (CC) of water upon examination.

3.8.11 Fording cutout. The alternator shall have a water sensing device such that when operated with a regulator so designed, will cause the alternator output to fall to (0) zero when the alternator is submerged per paragraph 4.5.13.1C, then regain normal output when withdrawn from the water.

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3.8.12 Fungus resistance. When tested as specified in paragraph 4.5.11, the alternator shall not support microbial growth that would adversely affect performance and shall subsequently meet the requirements of paragraph 3.8.1.

3.8.13 Corrosion resistance. After the alternator has been subjected to the test specified in 4.5.14, the generating system shall meet the rated output performance and the alternator shall show no degredation.

3.8.14 Electromagnetic compatibility.

a. When the suppression is part of the alternator: The alternator when tested as a subsystem per paragraph 4.5.15 shall conform to the EMI Requirements of MIL-STD-461 for tactical and special purpose vehicles.

b. When the suppression for the subsystem is part of the regulator: The alternator when tested as a component per paragraph 4.5 15 shall not exceed 125 db conducted noise at the alternator output connector; the radiated noise level shall not exceed the requirement for Electrical/Electro Mechanical equipments.

3.8.15 Sand and dust. The unit shall be capable of meeting the rated output performance curve of Drawing 10947517 following the tests specified in 4.5.12.

3.8.16 Painting. All exterior surfaces of the alternator, except electrical connections and the extended shaft, shall be painted in conformance with requirements on the applicable drawing.

3.9 Identification of product.

3.9.1 Marking. The alternator shall be marked in accordance with Standard MIL-STD-130.

3.9.2 Nameplate. The alternator shall have a nameplate attached to the alternator frame in a conspicuous location, so that plate may be easily read when generating system is installed in vehicle. The nameplate shall conform to Specification MIL-P-514. The nameplate shall contain the following information:

Alternator - 100 Amps 28 Volts.
Military Part Number - 10947517.
Maximum Rated Speed - 8,000 RPM.
Specification - MIL-A-62219.
Serial Number -
Manufacturer - Name or Identification Code.

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3.10 Workmanship. Workmanship shall be such as to assure a product free of burrs, scratches, sharp edges, chips, corrosion, scale and dirt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. The first article sample (see 3.1 and 6.3) shall consist of five specimens of the alternator. The specimens, properly marked with identifying information, shall be representative of the units proposed to be furnished to the Government. Unless otherwise specified, first article inspection shall be done by the manufacturer under Government surveillance and shall consist of examination for the defects specified in 4.3.2.2, and testing as specified in Table I.

Table I

Test	Paragraph Number	Sample 1	2	3	4	5
Rated output	4.5.2	x	x	x	x	x
Load transients	4.5.3	x	x			
Endurance	4.5.4	x				
High temperature	4.5.5		x			
Low temperature	4.5.6		x			
Shock	4.5.7					x
Overspeed	4.5.9			x		
Vibration	4.5.8			x		
Fungus resistance	4.5.11			x		
Sand and dust	4.5.12			x		
Submersion	4.5.13					x
Voltage transients	4.5.16				x	
Electromagnetic interference	4.5.15				x	
Corrosion resistance	4.5.14				x	
Fording cutout	4.5.13					x

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4.2.1 Failure. Failure of a test specimen to pass any specified examination or test, may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling for alternators.

4.3.1.1 Lot formation. A lot shall consist of all alternators, of one type or part number, from an identifiable production period, from one manufacturer, submitted at one time for quality conformance inspection.

4.3.1.2 Samples. Conformance examination shall be selected in accordance with MIL-STD-105.

4.3.2 Examination inspection.

4.3.2.1 Acceptable quality level. Each alternator selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be as classified in Table II.

Table II. Classification of defects

Categories	Defects
Major	
101	Dimensions affecting interchangeability not within tolerance (see 3.4)
Minor	
201	Dimensions not affecting interchangeability not within tolerance (see 3.4)
202	Improper finish (see 3.8.17).
203	Improper marking (see 3.9).
204	Poor workmanship (see 3.10)

4.3.3 Conformance inspection.

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4.3.3.1 100% quality conformance. All alternators furnished under this specification shall be subjected to operational tests of Table III except that the alternators shall be run in a system, including regulator, for one minute at room ambient. At the end of one minute, the output current shall not fall below the rated performance curve of Drawing 10947517.

Table III.

Test	Requirement	Test Condition
Output performance	3.8.1	4.5.2@1500 & 5000 RPM Pts
Speed balance		4.5.10
Overspeed	3.8.5	4.5.9

4.3.3.2 Rejection and retest. Alternators that have been rejected may be reworked or have parts replaced to correct defects and be resubmitted for examination and tests. Prior to resubmitting, detailed information on cause of rejection and the action taken to correct the defects shall be furnished to the Government inspector.

4.4 Control tests.

4.4.1 Sampling for control tests. Control test samples shall be selected at the rate of one alternator for each 250 produced, but not to exceed more than two in any 30 day period. Samples shall be from a production lot which has passed the quality conformance examination specified in 4.3.3.

Table IV. Control tests in sequence

Test	Test Paragraph No.	Requirement
Output performance	4.5.2	3.8.1
Over speed	4.5.9	3.8.5
Submersion	4.5.13.2	3.8.10
Fording cutout	4.5.13.2	3.8.11

4.4.3 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government, that the deficiencies revealed by the test have been corrected.

4.5 Test procedures.

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4.5.1 Test conditions. All tests performed herein shall be done as a system consisting of alternator with integral rectifier and regulator Drawing 10947439 and shall be conducted under general laboratory conditions acceptable to the Government. The ambient room temperature shall be maintained at 77° ± 15°F. The specimens shall be stabilized before being subjected to the tests. During all tests, the specimen shall remain at the set voltage unless otherwise specified.

4.5.1.1 Instrument tolerances. Measuring instrument tolerance for use in tests herein are as follows:

Voltmeter DC	± 0.25%
Ammeter DC	± 0.50%
Tachometer	1%
Thermocouple	1%

4.5.1.2 Test circuit batteries. Batteries used in connection with tests shall conform to MIL-B-11188. Test circuit batteries shall be considered fully charged when batteries, after having been stored at the ambient laboratory temperature, will accept a charging current of not more than 4.5 amperes at set voltage. Two 6TN 12-volt batteries, or equivalent, connected in series shall be used. In case of high voltage reading, battery specific gravity shall be checked, and if over 1300, the battery shall be adjusted such that the specific gravity is less than 1300 and the charging current does not exceed 4.5 amperes.

4.5.1.3 Frame temperature. Frame temperature shall be measured near the center of the stator lamination stack on the outside diameter surface.

4.5.1.4 Stabilized temperature. The generating system shall be considered temperature stabilized when operation under constant conditions fails to cause a change in frame temperature of 2°F in any 15 minute period after the stabilized temperature has been established, and shall remain at set voltage unless otherwise specified. The test shall not be less than one hour.

4.5.1.5 Output tests. In performing output tests on the generating system the following procedures shall be observed:

(a) Connect the alternator in a system consisting of alternator with integral rectifier, regulator and battery.
(b) Unless otherwise specified herein, the battery shall remain connected for all tests.
(c) Speed, frame temperature, system output current voltage, field voltage and current shall be recorded.
(d) The voltage regulator shall be adjusted to 28 ± 0.2 volts with the alternator running at 2700 revolutions per minute (RPM) and producing 25 amperes (amps). This shall be accomplished after the system temperature has been stabilized.

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(e) The generating system speed and load shall be adjusted at the maximum load for each point on the speed-load performance curve with voltage maintained at 28 volts.
(f) The temperature of the generating system shall be stabilized (see 4.5.1.4) at each point on the performance curve.

4.5.2 Rated output test. To determine conformance to 3.8.1, the alternator shall be tested in a system for:

(a) The minimum speed at which 50, 75, 100 amperes can be obtained.
(b) The maximum current that can be obtained at 1,500; 2,000; 3,500; 5,000 and 8,000 RPM

4.5.3 Load transients. To determine system conformance to paragraph 3.8.3.1 under simulated electric shock conditions, the system shall be operated at the following conditions with the loads and batteries rapidly applied and removed. Tests shall be conducted at room ambient, 225°F and minus 65°F.

RPM	AMPS
2750	5
2750	100
2750	0
2750	75
4000	0
4000	100
4000	0
6000	100
6000	0
6000	100

With the load and battery applied, the system output voltage shall stabilize at 28 ± 0.5 volts within 0.3 seconds of any load change and show no physical degradation.

4.5.4 Endurance. To determine conformance to 3.8.4, the generating system shall be subjected to the following test conducted at room temperature:

a. Set regulator at 28 ± 0.2 volts.
b. Full load: Adjusted for 100 ampere output.
c. Variable load: Cycle consists of 0, battery load, 50% and 100% of rated output; equal time at each value extended over a period of one cycle. The contactor will be opened and closed five times during each complete load change cycle, of approximately one hour.

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d. Operate the generating system for a total of 2000 hours in accordance with speed, duration and load conditions indicated in Table V recording frame temperature, ambient temperature, generating system output voltage and current, alternator field voltage and current.

Perform the rated output test upon completion of the endurance tests.

Table V. Endurance tests

RPM	Duration	Load Condition
2750	333 Hours	Full Load
8000	166 Hours	Variable Load
6000	166 Hours	Full Load
6000	335 Hours	Variable Load
2750	333 Hours	Variable Load
8000	166 Hours	Full Load
5000	166 Hours	Variable Load
5000	335 Hours	Full Load

4.5.5 High temperature. To determine conformance to 3.8.6, the generating system shall be operated at 2750 RPM. After stabilization, the load shall be adjusted to five amperes less than that load which will cause the system voltage to drop from the set voltage, but the unit output performance shall not be derated by more than 25 percent of paragraph 3.8.1 requirements during operation. This test shall be conducted at an ambient air temperature of 225°F. Air movement in the high temperature chamber shall be held to a minimum to prevent a temperature gradient within the chamber. Any cooling resulting from excess air motion, resulting from generating system fan shall not be considered in connection with evaluating conformance with the above requirement. Following 200 hours of operation subject the alternator to the rated output test at the 100 ampere load at room ambient.

4.5.6 Low temperature. To determine conformance to 3.8.7, the generating system shall be subjected to the following test:

(a) Prior to testing, the generating system shall be stored at a temperature of minus 80°F for a period of 24 hours.
(b) After removal for storage the temperature shall be stabilized at minus 65°F.
(c) The generating system shall then be operated at 2700 RPM, at set voltage with a 100 ampere load for a period of four hours.
(d) Upon completion, subject the alternator to the rated output tests at the 100 ampere load, at room ambient.

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4.5.7 Shock. To determine conformance to 3.8.8 the alternator shall be tested in accordance with test condition I Method 213 of MIL-STD-202.

4.5.8 Vibration. To determine conformance to paragraph 3.8.9, the generating system shall be tested in accordance with test condition D, Method 204 of MIL-STD-202. The following special conditions shall apply:

(a) The alternator shaft shall be rotated at 1750 ± 50 RPM during vibration under a no load condition.
(b) The alternator shall be mounted as in a normal vehicle installation.
(c) Frequency range shall be 10 to 600 Hertz per second.
(d) Maximum load along the longitudinal (shaft) axis of 10G applied at the vibration table.

4.5.9 Overspeed. Operate the generating system at 8000 RPM at full load until stabilized. When stabilized disconnect the resistive load, open field and increase the alternator speed to 10,000 RPM for 15 seconds. Unit shall conform to paragraph 3.8.5.

4.5.10 Balance-speed check. With alternator operating at 2750 RPM adjust load to maximum output with voltage set between 27.5 to 28.1 volts with the regulator in a full-on condition. The output shall be a minimum of 3 percent above the rated output at set RPM.

4.5.11 Fungus resistance. To determine conformance with 3.8.12, documentation shall be provided furnishing evidence that exposed materials of the alternator have been treated to resist fungus growths which would be detrimental to the performance of the alternator. In the absence of such documentation, the alternator shall be tested in accordance with Method 508 of MIL-STD-810 except that the test period shall be continuous 90 days. Upon removal from the cabinet, examination shall be accomplished by means of removable plates or plugs, without disassembly and without disturbing any fungus growths that have developed. Upon completion of the examination the generating system shall be operated for one hour, without load at 6000 RPM before being subjected to the additional testing specified in paragraph 3.8.12.

4.5.12 Sand and dust. To determine conformance to paragraph 3.8.16, the generating system shall be tested to the procedure I Sand and Dust Test of MIL-E-5272, the generating system position in the dust chamber shall be as in the intended use. The dust velocity shall be 2,300 ± 500 feet per minute. For ten minutes of each hour of the test period, the generating system shall be driven at a speed of 2450 ± 50 RPM.

4.5.13 Submersion.

4.5.13.1 First article. To determine conformance to 3.8.10, the generating system shall be stabilized by operating at 28 ± 0.2 volts at 2700 RPM and

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a load current of 100 amps. Stabilization shall be accomplished in the dry condition. The generating system shall then be subjected to the following test cycle.

(a) The generating system shall be driven in the dry condition for a period of 60 hours at 6000 ± 100 RPM and no load except the battery load.
(b) The water sensing device shall be such that when the alternator is submerged to a point when the drive belts touch water the regulator output goes to zero amperes.
(c) The alternator shaft shall then be rotated for a period of one hour while completely submerged in a 5 percent (by weight) salt solution at a speed of 1750 ± 50 RPM. The above cycle shall be performed a total of eight times. At the end of the first cycle with the unit rotation measure the system voltage and load while the unit is still submerged. The output shall be zero and system voltage equal to battery voltage.

4.5.13.2 Control test.

a. When the water sensing device is required on the alternator drawing, it shall be such that when the alternator is submerged to a point when the drive belts touch water the regulator output goes to zero amperes.
b. With the alternator submerged in a 5 percent salt solution rotate alternator shaft at 1750 ± 50 RPM for a period of eight hours. After removal from the salt solution, unit shall conform to requirements of paragraph 3.8.10.

4.5.14 Corrosion resistance. To determine conformance to 3.8.13, the generating system shall be tested for resistance to corrosion in accordance with Method 101 of MIL-STD-202, except the length of the test shall be 200 hours. During the test the alternator shaft will be rotated for a ten minute period out of each eight hour test interval. After the test the alternator shall be rotated for ten minutes at 1750 ± 50 RPM and no load before being subjected to the additional testing specified in 3.8.13.

4.5.15 Electromagnetic compatibility. The alternator when tested as a subsystem, consisting of alternator with integral rectifier, ordnance regulator #10947433 and shielded interconnecting cable, shall be done in accordance with MIL-STD-461 for tactical vehicles. When tested as a component the test methods of Electrical/Electro Mechanical equipments of MIL-STD-461 shall apply.

4.5.16 High voltage transients. The generating system shall be tested in such a manner that the maximum load is dumped five times at one second intervals under the following conditions:

a. 2000 RPM -65°F, room ambient and ± 225°F.
b. 6000 RPM -65°F, room ambient and ± 225°F.

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Under such conditions the transients shall not exceed the characteristics of 3.8.3.2.

4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of Section 5 and specifications referenced herein. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging data sheet for the desired level of protection specified (see 6.2).

6. NOTES

6.1 Intended use. Alternators covered by this specification are intended for use in military motor vehicles, and on other military equipment to produce direct-current electrical energy for vehicular and various other requirements such as lighting communication equipment and battery charging under extreme or unusual conditions of climate, weather, terrain or military service.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class (see 1.2).
(c) Applicable drawing (see 3.4).
(d) Selection and method and applicable level of preservation and packing and level of packing of referenced specification (see 5.1).

6.3 First article. With respect to products requiring first article testing, arregister will be maintained of products that have passed the complete series of first article requirements. After initially passing all of the first article requirements, and unless otherwise specified, paragraphs 4.5.4 (Endurance) and 4.5.11 (Fungus resistance) may be eliminated from subsequent first article tests for a period of five years on subsequent contracts provided the design, materials and manufacturing processes of the product has not changed. The subsequent contracts must run concurrent with the existing contract with no break in production and the contractor shall certify in writing that the design, materials and manufacturing processes are the same as when originally tested.

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Complete retesting will be required every five years. Copies of test reports covering the complete series of first article tests, conducted in accordance with paragraph 4.2 should be forwarded to U.S. Army Tank Automotive Command, ATTN: DRSTA-Q, so that a register may be maintained.

6.4 Definition.

6.4.1 Set voltage. As used in this specification the term "set voltage" is defined as the output voltage to which the generator system is adjusted. The value of the voltage where referenced herein is 28 ± 0.2 volts. After setting, the voltage reading shall be accurate to ± 0.2 volts.

6.4.2 Generating system. As used in this specification the term Generating System consists of one 100 ampere alternator-rectifier and a 100 ampere regulator, in one integrated unit or as two separate components.

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